Shayan Huda Chowdhury

Product Engineer, Full-Stack & AI Systems, Public Interest Technology

Website | LinkedIn | GitHub | Email | Phone

EDUCATION: Columbia University, NY: BA in Computer Science (AI/ML Focus) & Public Policy **GPA: 4.02, Dean's List, 2026**

Relevant Coursework: Neural Networks & Deep Learning, High Performance Machine Learning, Advanced Machine Learning Projects, Artificial Intelligence, LLM Foundations & Ethics, Continual Learning & Memory Models, Science & Policymaking, Computater Applications in Healthcare

TECHNICAL SKILLS

Languages: Python, R, JavaScript/TypeScript, MATLAB, Java, C++, C, CUDA, Rust, Dart, SQL

AI/ML: PyTorch, TensorFlow/Keras, transformers, vLLM, LoRA/PEFT, PyTorch profiling, multi-model orchestration, agentic systems, RAG (keyword/BM25 + semantic), reinforcement learning, tool use, structured outputs w/ LLMs (open/closed-source)

Full-Stack: FastAPI, Next.js/React, SSE streaming, Databases (PostgreSQL, pgvector)

Infrastructure & DevOps: AWS (EC2, CI/CD), Docker, Nginx, PM2, Linux, zero-downtime deployments

TECHNICAL EXPERIENCE

Chief Technology Officer, Digital Transformation Global (DXG) Foundation, Toronto, Canada **Aug 2025 – present**

- Leading dev of **AI policy** platform at **Gates Foundation-funded** nonprofit, used by UNDP and government agencies across Global South nations including Bangladesh, Pakistan, and Tanzania for policy analysis and implementation across domains such as poverty alleviation
- Architecting **agentic AI systems + harnesses**, combining multi-hop **RAG** (semantic + keyword), OS-inspired **memory/context management** (retrieval/eviction policies), tool selection optimization, sandboxed code execution, and hallucination-prevention measures
- Built **full-stack platform** integrating LLM APIs, multimodal info extraction, HITL eval frameworks, and production-ready CI/CD pipelines

Executive Director & Founder, Reach4Help, Montréal, Canada & New York **Mar 2020 – present**

- **Leading a team of 30+** in coordinating **humanitarian aid logistics for disaster response in over 38 countries**, working closely with UN agencies, governments, and civil society organizations to ensure that aid reaches remote, vulnerable, and hard-to-access areas
- Developed an **open-source global map of aid**, serving as the backbone for logistical coordination of volunteer help for **10K+organizations** through the COVID-19 pandemic, invasion of Ukraine, wildfires in the US, earthquakes in Mexico, floods in South Asia, and more
- **Raised $120K** from Google for Nonprofits and **secured $106K** in in-kind donations from Google, Slack, Algolia, etc.
- **Helped raise €170K+** for medical relief and civilian defense in Ukraine + **provided aid for 58K+ households** during floods in Bangladesh, Pakistan, and India, partnering with the World Economic Forum and NGOs such as BRAC, Kandari, etc.
- Built and **deployed HeyNYC,** a civic services assistant for New Yorkers to navigate **government benefits**, housing, food, health, cooling locations, emergency alerts, and events using official NYC APIs, agency sources, and NYC Open Data with privacy and citation safeguards

Research Assistant, Artificial Intelligence in Medicine Program, Harvard Medical School, Boston, MA **Jun 2024 – present**

- Developing **LLM-based methods** for **cancer risk assessment and triage** at Dana Farber Cancer Institute / Harvard Med to inform public health decision-making in clinical settings, **advised by Dr. Danielle Bitterman** in the Radiation Oncology department
- **First authored** paper accepted to **ML4Health** (**NeurIPS 2024**) on **reinforcement learning (RL)** methods for **synthetic data generation**; demonstrated how **open source LLMs** (Mistral-7B) fine-tuned on **synthetic patient data** generated from <2% of available scarce clinician-labeled data **outperform fully-supervised baselines** (LLMs trained on entire clinician-labeled dataset) and GPT-4 few-shot
- Developing **LLM-based medical information extraction** systems for **pediatric cancer patients** across 200+ clinical notes; achieved 2.6× improvement over clinical NLP baselines; w/ funding from the **National Cancer Institute (NCI)**, pending submission to **JAMA Oncology**

Undergraduate Researcher, Department of Computer Science, Columbia University, New York **Sep 2023 – Jun 2026**

- Co-developed methods for **dynamic KV cache compression** for LLM reasoning (*under review*, **ACL 2026 SRW**) + identified "entropy neurons" in Gemma models for **mechanistic interpretability of LLM uncertainty** (*under review*, **ICML 2026**) w/ **IBM Research**
- Designed and implemented **end-to-end NLP pipeline** for reasoning, question generation, multi-source retrieval, and argumentation using open-source LLMs for **fact-checking**; **first-authored** paper to **FEVER workshop** at **ACL 2025,** advised by **Prof. Smaranda Muresan**
- Developed **knowledge graphs** to **streamline drug discovery** by embedding pharmaceutical ontologies using NLP tools like SUSIE for entity extraction, enabling structured querying and insights for **drug development**, advised by **Prof. Venkat Venkatasubramanian**
- Developed **ML classifiers** to **identify foodborne illness** hotspots across **~16K+ restaurants** w/ 10 years of data (2012-2021) in collab w/ **NYC Department of Health** to inform public health policy on outbreak prediction and prevention; **second authored paper** on findings presented at the **2024 International Conference on Social Media & Society** in **London, UK**, advised by **Prof. Luis Gravano**

Lead Data Analyst, a2i Programme, ICT Ministry / Cabinet Division / UNDP Bangladesh **Mar 2020 – Jan 2022**

- Built **data processing infrastructure** handling **21M+ surveillance reports,** patient data from **600+ hospitals,** enabling near-real-time analytics across **171M people** for **national public policy interventions**; **led presentations** to head policymakers **up to the Prime Minister**
- Developed **statistical models** using data on tests, cases, deaths, hospitalizations, and mobility for **epidemiological forecasting**; **co-authored papers** in Nature Comms and Epidemics alongside MIT and Cambridge faculty on **disease dynamics** and **policy response**
- **Appointed** to the National COVID-19 Policy Dashboard Committee of the Bangladesh Government for the **Ministries of Health and IT**

Research Technician, Fifer Lab, Columbia University Irving Medical Center, New York **Aug 2021 – Sep 2023**

- Developed **a signal processing algorithm** to extract fetal **ECG** data from 50+ high-risk pregnant mothers in South Africa, resulting in **enhanced monitoring** and **early intervention** of neurodevelopmental disorders in newborns; **first-authored paper** to IEEE EMBC 2022

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- Built **self-supervised ML models** for **real-time clinical analysis** of fetal + maternal ECG data, towards **improving access to low-cost technologies** for **preventing fetal miscarriages and stillbirths**, w/ focus on clinical settings in **low+middle-income countries (LMICs)**

Software Developer, Migrant Nation Foundation, The Netherlands / UNDP Bangladesh **Oct 2019 – Jun 2020**

- **Developed an e-commerce marketplace** for **11K+ Rohingya refugees** in Bangladesh, enabling them to **earn a living wage** and **achieve financial independence** by producing and selling products to consumers worldwide on Amazon, Alibaba, and other international markets
- **Lived and worked in the camps** with several United Nations agencies including UN Development Programme (UNDP), UN High Commissioner for Refugees (UNHCR), and the World Food Programme (WFP)
- **Built a mobile app to pay refugees hourly wages** in accordance with the International Labour Organization's decent work practices

Bioinformatics Research Intern, Frank Lab, Columbia University, New York **Sep 2018 – Nov 2019**

- Developed **noise-reduction algorithms for cryo-EM images** to achieve the **highest resolution 3D reconstruction** of the 40S eukaryotic ribosome subunit (at ≈0.15 nm), reverse-engineering protein synthesis and cancer determinants, under **Nobel Laureate Joachim Frank**
- **Fine-tuned bioinformatics tools** to increase the accuracy of protein reconstruction in noisy electron microscopy images

Machine Learning Project Lead, Google Mentorship Program, Google, New York **Oct 2018 – Jun 2019**

- Led development and training of **LSTM** and **Transformer models** with Keras & PyTorch for **voice emotion detection** from speech audio, targeting applications in **Google Assistant**, **clinical diagnostics**, and **accessibility for speech disabilities**

SELECTED RESEARCH PUBLICATIONS (GOOGLE SCHOLAR FOR FULL LIST)

- **S Chowdhury**, S Fang, S Muresan (2025). FACT5: A Novel [LLM] Benchmark and Pipeline for Nuanced Fact-Checking of Complex Statements. *Proceedings of the Eighth Fact Extraction and VERification Workshop* (FEVER), ACL 2025.
- **S Chowdhury**, MG Frasch, M Lucchini, LC Shuffrey, A Sania, C Malette, HJ Odendaal, MM Myers, WP Fifer, N Pini (2022). A Novel Method for the Extraction of Fetal ECG Signals from Wearable Devices. *44th Int'l IEEE* Engineering in *Medicine & Bio Conference (EMBC)*.
- AS Mahmud, **S Chowdhury**, KH Sojib, A Chowdhury, MT Quader, S Paul, MS Saidy, R Uddin, K Engø-Monsen, CO Buckee (2021). Participatory syndromic surveillance as a tool for tracking COVID-19 in Bangladesh. *Epidemics* 35, 100462.
- F Chadwick, J Clark, **S Chowdhury**, … & Sania, A. (2022). Combining Rapid Antigen Testing and Syndromic Surveillance Improves Sensitivity and Specificity of COVID-19 Detection: A Community-Based Prospective Diagnostic Study. *Nature Communications*, 13, 2877.
- EA Ferguson, E Brum, A Chowdhury, **S Chowdhury,** M Kundegorski, … & K Hampson. (2022). Modelling how face masks and symptoms-based quarantine synergistically and cost-effectively reduce SARS-CoV-2 transmission in Bangladesh. *Epidemics*, 40, 100592.

SOCIAL ENTREPRENEURSHIP EXPERIENCE

Global Shaper, Curator & Climate Reality Leader, World Economic Forum, Geneva, Switzerland **Aug 2021 – present**

- **Selected as the youngest Global Shaper** by the World Economic Forum for work on Reach4Help
- **Led SupportUkraineNow projects** as U.S. co-ambassador, reaching **1M+ Ukrainians** on refugee aid, fundraising and mental health support
- **Awarded a grant from Al Gore's Climate Reality Project** to scale **education policy solutions** with the Dhaka Hub in Bangladesh, impacting **900+ students** through climate action and education initiatives: **plastic recycling, tree planting, and hands-on workshops**
- **Elected to be Curator of the New York City Queens Hub** for global representation and local leadership, focusing on **youth education and in technology and policy** through partnerships with local NGOs and tech companies

SELECTED HONORS & AWARDS

- NYC Queens Hub Curator, Global Shapers, World Economic Forum **2024**
- Climate Reality Corp Leadership Award, awarded by fmr. U.S. Vice President Al Gore **2022**
- Dean's List (All Semesters), Columbia University **2022–**
- 5 Scholarships Awarded By Columbia University: Distinguished New Student Scholarship, Elizabeth Piper Scholarship, etc. **2018–**

SELECTED SPEAKING ENGAGEMENTS, EVENTS & PRESS COVERAGE

- Lead author, **"FACT5: A Novel Fact-Checking Benchmark + Pipeline"** at the **8th FEVER Workshop** at **ACL 2025** in **Vienna, Austria** **Jul 2025**
- Speaker, **ASU+GSV (Global Silicon Valley) Summit 2025** invited by the **Aspen Institute** in **San Diego, California** **Apr 2025**
- Curator, **Global Shapers Annual Summit 2024** organized by the **World Economic Forum** in **Geneva, Switzerland** **Jul 2024**
- Guest Lecturer, **What Was AI Made For** at **Columbia University Double Discovery Center** Summer Academy in **New York** **Jun 2024**
- Panelist, **"Bringing Health and Equity into Climate Action"** organized by **Accenture** and **World Economic Forum** in **New York** **Sep 2023**
- Lead presenter and author, **International Society for Developmental Psychobiology (ISDP) Conference** in **San Diego, California** **Nov 2022**
- Presenter, **"The Roadmap to Resources: How Reach4Help Leads People in Need"** at **Google HQ** in **Mountain View, California** **Oct 2022**
- Speaker, **Climate Changemakers Digital Innovation for Youth Conference** at the **United Nations Headquarters** in **New York** **Sep 2022**
- Lead presenter and first author, **44th Int'l IEEE Engineering in Medicine & Biology Conference (EMBC)** in **Glasgow, Scotland** **Jul 2022**
- Exclusive feature in **The Daily Star**, the most widely circulated English newspaper in **Bangladesh** **Mar 2022**

SPOKEN LANGUAGES: Bangla (native), Mandarin (int.), Spanish (int.), Latin (studied)